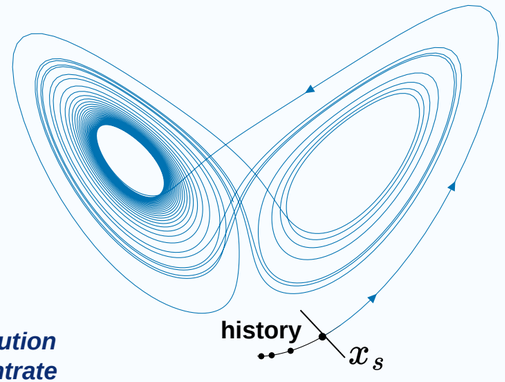
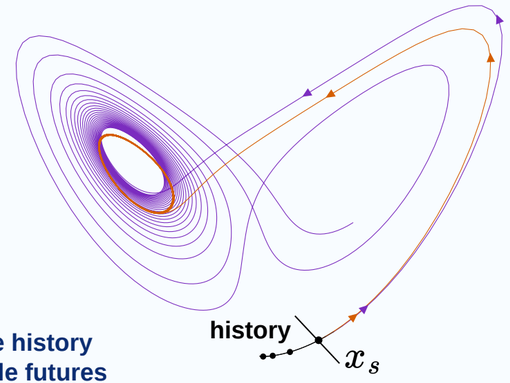
A. Physical-Time Flow Forecasting

Fully Observed Deterministic Dynamics



Distribution concentrate

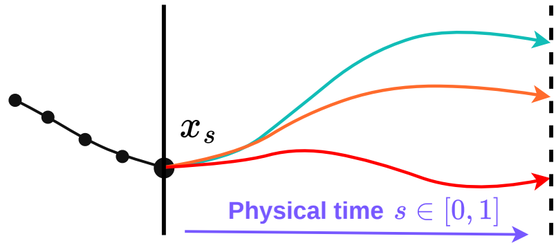
Partial Observed Stochastic Dynamics



Same history multiple futures

Latent Conditioned Velocity

$$\frac{d\hat{x}_s^{(j)}}{ds} = v_\theta\left(\hat{x}_s^{(j)}, s, h, z^{(j)}\right), \quad z^{(j)} \sim p(z)$$



B. VSF in LeWM Planning

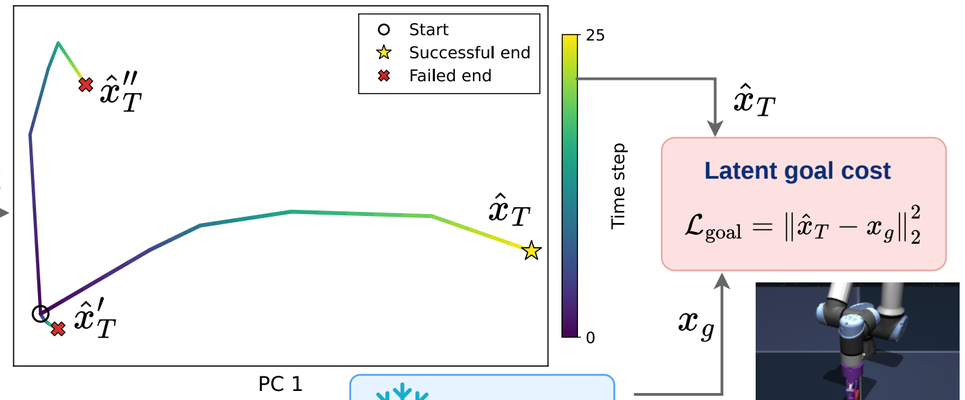
CEM candidate actions

A^1 ⋯
A^2 ⋯
A^3 ⋯

Action-conditioned VSF predictor
replaces AR predictor

$$\frac{d\hat{x}_s}{ds} = v_\theta\left(\hat{x}_s, s, h, a_{m+k}, z\right)$$

VSF representation trajectory



Latent goal cost

$$\mathcal{L}_{\text{goal}} = \|\hat{x}_T - x_g\|_2^2$$



o_s

$\hat{x}_s$

Encoder

shared weights

Encoder

x_g

o_g

